

KLS Gogte Institute of Technology, Belagavi

IA Test – I

Program: B.E

Academic Year: 2024-25 (ODD SEM)

Department: CSE

Semester: III

Max. Marks: 30 marks

Course Title: Object Oriented Programming using Java

Code: 22CS33

Duration: 1 Hr.

Date: 25-09-2024

Instructions:	1	Missing data, if any, may be suitable assumed.
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Q.No	PART A (Compulsory question)	[L]	[CO]	[PO]	[M]
1.	a) What are the Program outcomes that are mapped to “ Explaining the concept of encapsulation in Java with a code snippet demonstrating getter and setter methods. ” b) Distinguish between a class and an object? Give the general form to create them with suitable example . c) Explain the syntax of for – each style for loop with suitable example	2	1	1,2	2 4 4
PART B (Answer any 1 questions)					
2.	Two workers, Ram and Sham, are working on separate parts of the building’s foundation, and they need to calculate the total distance covered by combining their individual measurements. Develop a Java program that helps the workers calculate the total combined distance between their two sections of work. The distances measured by Ram and Sham are given in feet and inches. Design a Distance class having feet and inches as instance variables. include the following in the class. i) Default constructor. ii) Parameterized constructor to initialize the instance variables iii) A method that add two distance object and returns a resulting distance object	3	1	1,2,3	10
3.	Imagine you're a software developer tasked with creating a program for a geometry tutoring app. The app helps students learn about different types of triangles and their properties by allowing them to input the lengths of the sides of a triangle and receiving feedback on the type of triangle it is (Equilateral, Isosceles, or Scalene) and its area. Design a class named myTriangle that models a triangle using its three sides. The class should have the following functionalities: a) Initialization of the Triangle b) Determine the Type of Triangle c) Compute and Return the Area	3	1	1,2,3	10

P.T.O

PART C (Answer any 1 question)

4.	Analyse the following program and write its output <pre> public class StringHandling { public static void main(String[] args) { String str = "Hello KLSGIT students!"; String str1 = "Hello"; String str2 = "Hello"; String str3 = new String("Hello"); System.out.println(str.length()); System.out.println(str.charAt(6)); System.out.println(str.substring(7, 11)); char[] dst = new char[6]; str.getChars(6, 12, dst, 0); System.out.println(dst); String str4 = "Let's start coding!"; System.out.println(str.concat(str4)); System.out.println(str1.equals(str2)); System.out.println(str1.equals(str3)); System.out.println(str1.compareTo(str2)); System.out.println((str1 == str3)); } } </pre>	4	1	1,2	10
5.	A weather station collects temperature data every hour throughout the day. A scientist wants to analyse this temperature data to determine the hottest, and coldest hours, the average temperatures of the day and the number of hours where the temperature exceeded the threshold of 30 degrees Celsius. Analyze this scenario and implement the same using Java.	4	1	1,2,3	10

IA Test – II

Program: B.E

Academic Year: 2024-25 (ODD SEM)

Department: CSE

Semester: III

Max. Marks: 30 marks

Course Title: Object-Oriented Programming using Java

Code: 22CS33

Duration: 1 Hr.

Date: 13-11-2024

Instructions:	1	Missing data, if any, may be suitable assumed.
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Q.No	PART A (Compulsory question)	[L]	[CO]	[PO]	[M]
M3 1.	a) "Develop a class hierarchy for a vehicle management system that handles different types of vehicles (e.g., Car, Truck, Bike). Each vehicle has common properties such as make, model, and year, and they all have a method startEngine() to start the vehicle." Define the COs and POs that are appropriate to this question	2			2
MM	b) How do Packages participate in access control? Explain with an example	2	4	1	4
MM	c) Explain the different ways in which we can use the 'final' keyword with an example	2	4	1	4
PART B (Answer any 1 question)					
M3 2.	Construct a Java program that accepts two integer arrays from the user. The program should perform the following steps within a nested try-catch structure: Inner try block should divide the first array elements by the second array elements and handle any attempt to divide the first array by 0. Outer try block should handle array index range Regardless of whether exceptions occur or not, the program should display a message indicating that the program has finished processing. Write an appropriate code to handle both successful execution and the scenarios when exceptions are thrown.	3	1, 2	1,2,3	10
M3 3.	A transportation app supports multiple modes of transport such as cars, buses, bicycles, and scooters. Each mode has different costs, routes, and estimated travel times. The app needs to dynamically calculate the best travel mode based on cost and time. Apply the interface concept to build the application	3	1, 4	1,2,3	10
PART C (Answer any 1 question)					
M3	Design a library system where there are multiple types of media items, such as Books, Magazines, DVDs, and Newspapers. Each of the media items has common properties such as title, year published, and available copies. Each media type also has its own unique properties (e.g., Books might have author and genre, while DVDs might have duration and director etc.). The	4	1	1,2,3	10

	system should have functionalities for checking out, returning the item, and creating a report function that lists all available media items with their specific details. Analyze how inheritance allows each media type to share common attributes and behaviours while still supporting unique properties.			
M/5.	When you are developing a payment processing module for an e-commerce application, if a user inputs incorrect card details such as an expired card, incorrect CVV, or invalid card number, handle these cases by throwing specific exceptions like InvalidCardException, ExpiredCardException, or InvalidCVVException. Requirements: • Create a method processPayment(String cardNumber, String cvv, String expiryDate). • If any card details are invalid, throw the respective exceptions. • Catch these exceptions in the main method and provide detailed error messages prompting users to correct their input. Analyze these requirements to build an appropriate system	4	2	1,2,3

OBJECT ORIENTED PROGRAMMING USING JAVA

Time: 3 hrs.

Max. Marks :100

Instructions:

1. Part A: Answer any FIVE questions and each question carries 6 marks.
2. Part B: Answer any ONE full question from each module and each question carries 10 marks.
3. Part C: Answer any ONE full question and each question carries 20 marks.

PART A**Instructions:** 1. Answer any FIVE Full Questions.

	L	CO	PO	M
✓ 1. Explain the three key attributes of Object-Oriented Programming.	[2]	[1]	[1]	[6]
✓ 2. Explain class and object w.r.t object oriented programming with syntax and example.	[2]	[1]	[1]	[6]
✓ 3. Explain the two ways of using super keyword in inheritance.	[2]	[2]	[1]	[6]
✓ 4. Explain how to achieve multiple inheritance in Java.	[2]	[2]	[1]	[6]
✓ 5. Define exception handling. Explain the 5 keywords used in exception handling.	[2]	[3]	[1]	[6]
6. Explain static import of packages with a suitable example.	[2]	[5]	[1]	[6]
7. Explain the collection interfaces packaged in java.util.	[2]	[4]	[1]	[6]

PART B**Instructions:** 1. Answer any FIVE Full Questions selecting at least ONE Question from Each Module.**MODULE 1**

8. Build a seat reservation system for a cinema. The cinema hall's seating arrangement is represented by a 2D grid, and each seat can either be available or reserved.

Add the following functionalities:

- a) Prompt the user to enter the number of rows and columns to set up the seating arrangement.
- b) Take user input to mark some seats as reserved or available.
- c) Prompt the user for the row and column of a seat they want to reserve.
- d) If the seat is available, mark it as reserved and print the updated seating chart.
- e) If the seat is already reserved, display a message saying it is taken.

[3] [1] [1,2,3] [10]

OR

9. Construct an electronic application in Java which has a class Fan that has an attribute to represent if it's on or off, with methods to turn the fan on or off, and check the status.

[3] [1] [1,2,3] [10]

MODULE 2

10. Develop a Java program that contains an employee class with attributes like name, id, and salary. Add the following functionalities:
- Take an employee's object as the parameter and compare his salary with another employee's salary. Display the appropriate message.
 - Show the details of each employee.

[3] [1] [1,2,3] [10]

OR

11. Build a Java program which has a class called SearchClass with three overloaded methods named search. Each method should be designed to search for a specific data type - integer, float, and String. Each search method will take an array and a key of the respective type, look for the key in the array, and return its index if found, or -1 if not.

[3] [1] [1,2,3] [10]

MODULE 3

12. Develop a banking system to handle different types of accounts with specific transaction rules. Define the following classes:

- Bank Account:** A base class with a withdraw() method that deducts a specified amount from the account balance.
- Savings Account, Current Account:** Each subclass overrides withdraw() with specific withdrawal rules:
 - Savings Account:** Allows withdrawals only up to a certain limit per day.
 - Current Account:** Allows overdraft up to a specified limit.

[3] [1] [1,2,3] [10]

OR

13. Construct an Arithmetic Series Generator using an interface Series with the following methods:

- int getNext(): Returns the next term in the arithmetic series.
- void reset(): Resets the series to its starting value.
- void setStart(int start): Sets the initial term of the series.

Implement the ArithmeticSeries class that generates terms by adding a fixed difference to the previous term. The program should allow the user to set the starting value, and common difference, generate terms, and reset the series.

[3] [1,4] [1,2,3] [10]

MODULE 4

14. Build a flight booking system where users can book seats on flights with the following functionalities:

- Implement a method bookSeat(int flightId) that checks if seats are available on a specified flight.
- If no seats are available, throw NoSeatsAvailableException.
- Catch this exception and inform the user, suggesting alternative flights if available.

[3] [2] [1,2,3] [10]

OR

15. Build a package named employees with a class Employee having private, default, protected, and public members. Create another class in the same package that tries to access these members and demonstrate the visibility restrictions.

[3] [4] [1,2,3] [10]

MODULE 5

16. Develop a financial software system to calculate taxes for employees based on their income. Implement a TaxCalculator interface with a method calculateTax(double income) to apply different tax rates based on income brackets. Use a Predicate to check if the calculated tax is below 1000. Process the incomes of 5 employees, compute the tax, and check if the tax is less than 1000 using Lambda expression. Print the tax amount and the result of the check for each employee.

[3] [3] [1,2,3] [10]

OR

17. Build a Java program that implements an ArrayList of Strings using the Collection framework. Perform the following tasks on the ArrayList:

- Add the following entries to the ArrayList:
["Alpha", "Beta", "Gamma", "Delta", "Epsilon", "Zeta", "Eta"]
- Use an Iterator to:
 - Display the contents of the list.
 - Remove the element "Gamma" from the list and display the contents after deletion.
- Use a ListIterator to:
 - Add "Gamma" back to the list.
 - Modify the elements being iterated, and display the list backwards.
- Sort and display the ArrayList elements in both ascending and descending order.

[3] [3] [1,2,3] [10]

PART C

Instructions: 1. Answer any ONE full question.

L CO PO M

- 18.a. Orion company, manufacturer of electric vehicles has branches in 4 zones of India namely North, South, East and West. The company wants to find the following things:

- Total sales from Jan to Dec in each zone
- The highest sale zone in each month from Jan to Dec
- Zone with highest average sales

Analyse the scenario and solve this using 2-dimensional array where the rows represent the zones and columns represent the months from Jan to Dec. The values in array are the number of sales in a particular zone and month.

[4] [2] [3] [10]

- 18.b. Analyse the given scenario and develop a java program. A private library issues 3 books to its regular members and 8 books to goldcard members. The maximum duration of lending books for regular members is 7 days and for goldcard members 14 days. Implement a base class 'Member' with fields as name, Id, borrowedBooks and duration. Use constructor to initialize the fields. Declare an abstract method borrow(int count, int duration) which should be implemented in subclasses. Implement RegularMember and GoldcardMember as subclasses of class Member. The borrow() method should throw BorrowingLimitExceededException and DurationExceededException whenever necessary. The program should handle the given custom exceptions properly.
- [4] [3] [3] [10]

OR

- 19.a. Analyse the given scenario and develop a java program. A fashion designer's website has announced a sale and it wants an automated system to compute the bill. If a customer buys one outfit, 10% discount should be given on the price. For two outfits, 15% discount and for 3 outfits, 20%. Use the concept of method overloading to compute the bill by applying appropriate discount based the number of outfits bought. Pass the price of the outfits as the parameters for the overloaded method computeBill().
- [4] [2] [3] [10]
- 19.b. Analyse the given scenario and develop a java program. A teaching faculty wants to maintain the data of mentees such as name, year_of_admission, current_sem, CGPA and no_of_courses_registered etc. If the student is regular, all the fields should be initialized with appropriate values. If the student is non-eligible, then only name and year_of_admission should be initialized with proper values and remaining fields should be initialized with null/zero. Use constructors for initialization. Define a display() method to display all the information of students.
- [4] [2] [3] [10]

USN : 2GT23CS157

COURSE CODE : 22CS33 / 22IS33

THIRD Semester B.E. Semester End Examination, JUNE 2025

OBJECT ORIENTED PROGRAMMING USING JAVA

Time: 3 Hours.

Maximum Marks :100

Instructions:

1. Part A: Answer any FIVE questions and each question carries 6 marks.
2. Part B: Answer any ONE full question from each module and each question carries 10 marks.
3. Part C: Answer any ONE full question and each question carries 20 marks.

PART A

Instructions: 1. Answer any FIVE Full Questions.

	L	CO	PO	M
1. Distinguish between an abstract class and an interface.	[2]	[5]	[1]	[6]
2. Summarize the types of constructors with an example for each.	[2]	[2]	[1]	[6]
3. List and explain the properties of Object-Oriented Programming.	[2]	[1]	[1]	[6]
4. Explain the consequences of an uncaught exception with an example.	[2]	[3]	[1]	[6]
5. Illustrate different types of lambda expressions with an example for each.	[2]	[4]	[1]	[6]
6. Explain the various components of Java Collection Frameworks with examples.	[2]	[3]	[1]	[6]
7. Distinguish between a class and an object.	[2]	[1]	[1]	[6]

PART B

Instructions: 1. Answer any FIVE Full Questions selecting at least ONE Question from Each Module.

MODULE 1

8. Develop a hotel room booking system with the following functionalities:
- a) Implement a method bookRoom(int roomId) to check if a specific room is available.
 - b) If the room is unavailable, throw a RoomUnavailableException.
 - c) Catch the exception and provide the user with a list of alternative available rooms, if any.
- [3] [1] [1,2,3] [10]

OR

9. Develop a Java program to find the average score of five students in three papers.
- [3] [1] [1,2,3] [10]

10. Implement a Java program to represent a Complex number and include the following members:

- Default constructor to initialize a complex number to 0
- Parameterized constructor to initialize a complex number to a user defined value
- Method which takes 2 complex numbers as the parameters, adds them, and returns the result
- Method to subtract two complex numbers and return the result.
- Display a complex number

[3] [1] [1,2,3] [10]

OR

11. Develop a class "SortingClass" that has three overloaded methods to sort data of int, float, and String types. Demonstrate its working by passing an array of all the above three types. Use any sorting technique.

[3] [1] [1,2,3] [10]

MODULE 3

12. Solve the following by creating a class Manager with attributes: name, ID and basic salary. Classes HR_Manager and Sales_Manager inherit from Manager and compute gross salary as per the following:

	DA	HRA	TA	Deductions	Gross Salary
HR Manager	70% of basic	20% of basic	NA	5% of basic	DA+HRA-deductions
Sales Manager	70% of basic	10% of basic	= 5% of basic	5% of basic.	DA+HRA+TA-deductions

Demonstrate how runtime polymorphism is achieved.

[3] [1] [2,3] [10]

OR

13. Implement a Robot that declares three methods start(), work(), and stop(). Create a Disposal class that implements the methods in the Robot and adds the findWaste() method which returns true if the waste is wet, false otherwise. Write the associated driver class.

[3] [4] [1,2,3] [10]

MODULE 4

14. Develop a Java program that takes the value from a user and passes it into a method that checks whether it is odd, then throws a user-defined exception OddNumberException if it is else throws EvenNumber Exception. Implement user-defined exceptions and handle these exceptions carefully.

[3] [2] [1,2,3] [10]

OR

15. Develop a Calculator class with methods add, subtract, multiply, and divide. Each method should take two parameters (operands) and perform the corresponding arithmetic operation. Implement exception handling to catch and handle potential exceptions, such as ArithmeticException for division by zero or NumberFormatException for invalid input.

[3] [2] [1,2,3] [10]

MODULE 5

16. Construct a program that counts the frequency of each element in a collection and displays the results.

[3] [3] [1,3] [10]

OR

17. Build a program that utilizes lambda expressions to sort a list of custom objects based on different criteria (e.g., age, name, salary).

[3] [3] [1,3] [10]

PART C

Instructions: 1. Answer any ONE full question.

L CO PO M

- 18a. Create a music player system using interfaces. Design two interfaces: MusicPlayer and Playlist. The MusicPlayer interface should have methods for playing, pausing, and stopping music, while the Playlist interface should have methods for adding and removing songs from a playlist. Implement a class MyMusicPlayer that implements both interfaces. Use the interfaces to ensure that the music player supports basic playback functionality and playlist management. Explain the use of the music player by creating an instance of MyMusicPlayer and interacting with it.

[4] [4] [1,2,3] [10]

- 18b. Imagine an e-commerce platform like Amazon, where customers can shop for a wide range of items, including electronics, clothing, groceries, and cosmetics. Each of these categories has specific details that need to be captured, such as brand and warranty for electronics, size and material for clothing, weight and expiration date for groceries, and brand and type for cosmetics. To efficiently manage the addition of items to the shopping cart, the platform's backend system uses method overloading to handle each product category individually, ensuring that the right information is captured for each type of item. Analyze and develop a Java program to build the system

[4] [1] [1,2,3] [10]

OR

- 19a. Ronak wants to develop an app for his movie theatre that manages seat reservations to help his villagers book tickets. The theatre has a limited number of seats, and customers can choose and reserve seats based on availability. The program should display a seating chart, allow seat reservations, and provide an option to cancel the application once the customer is done.

[4] [1] [1,2,3] [10]

- 19b. You are part of the development team for a new word game called "Word Shuffle". In this game, players must match pairs of words that are anagrams of each other to earn points. An anagram is a word that can be formed by rearranging the letters of another word, such as "listen" and "silent." Your task is to implement this game to check if two given words are anagrams. This feature will allow players to input two words, and the program will output whether or not they are anagrams, helping players verify their answers during the game. Analyse the game rules and implement the same.

[4] [1] [2,3] [10]